



Recall that a *random variable* is a variable whose possible values are numerical outcomes of a random phenomenon such as a die-throwing experiment, and that *discrete* random variables are quantities that take on a *countable* number of possible random values, for instance the number of emails that one can receive a day can be 0, 1, 2, 3, In this topic, we will learn a new type of random variable, called a continuous random variable.

§1 Continuous Random Variables

Key Questions:

- ☐ What is a *continuous random variable*?
 - ☐ How is it different from a discrete random variable?
 - ☐ What is a *probability density function*? How is it used to calculate probabilities for a continuous random variable?
 - ☐ What is the total area under a *probability density curve*?

What is a continuous random variable?

Definition 1.1 (Continuous Random Variable)

A random variable whose possible values form a continuous interval, such as $[0, 1]$, or even the entire set of real numbers, $\mathbb{R}$, is called a *continuous random variable*.

In comparison to discrete random variables, continuous random variables differ in that they are quantities that take on *uncountably* many possible random values, and thus are described by an *interval* of values.

Some examples of continuous random variable are as follows:

Random Variable	Possible Range of Values
Length of one part cut from a random point along a 10-cm string	(0 cm, 10 cm)
Height of a randomly selected 18-year old male student	[1.2 m, 2.4 m]
Weight of a randomly selected 18-year old female student	[30 kg, 90 kg]

How do we determine whether a random variable is discrete or continuous?

Example 1.2

Determine if the following random variables are discrete or continuous.

Random Variable	Discrete / Continuous?
Number of students who come to college by bus each day	Discrete
Sugar concentration in a randomly chosen orange	Continuous
Length of time a randomly chosen student spends studying each week	Continuous
Number of tails obtained when three coins are tossed once	Discrete

Recall that a discrete random variable is described by its probability *distribution* function (or equivalently probability mass function). For example, for a binomial random variable $X \sim B(n, p)$, its probability *distribution* function is

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

For a continuous random variable, the probability distribution is instead described by a *probability density function*, which is defined as follows:

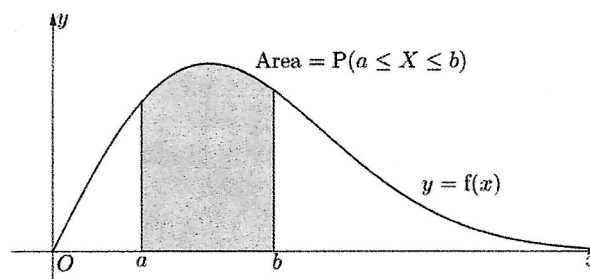
What is the probability density function of a continuous random variable?

Definition 1.3 (Probability Density Function)

For a **continuous** random variable X , its *probability density function*, $f(x)$, satisfies

$$P(a \leq X \leq b) = \int_a^b f(x) \, dx$$

for any possible values a and b of X .



The curve with equation $y = f(x)$ is called the *probability density curve* of X .

How is it used to calculate probabilities for a continuous random variable?

In other words, the area under the probability density curve for a specific interval gives the probability that X takes on values in that interval.

The following are some key properties of all probability density functions:

What are the properties of a probability density function?

Properties of a probability density function, $f(x)$

Non-Negativity

The probability density function is non-negative at all values of x , i.e.,

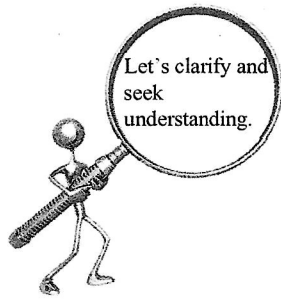
$$f(x) \geq 0 \text{ for all } x.$$

Probability of Sample Space is 1

The total area under the probability density curve is 1, i.e.

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

Do note that $f(a)$ is **NOT** the probability that $X = a$, i.e. $P(X = a) \neq f(a)$.



For any continuous random variable X and value a , what is the value of $P(X = a)$?

From Definition 1.3,

$$\begin{aligned} P(X = a) &= \int_a^a f(x) dx \\ &= 0. \end{aligned}$$

§2 Normal Distribution

Key Questions:

- ☐ What is the *standard normal random variable*?
 - ☐ What is the *mean* and *variance* of the standard normal random variable?
 - ☐ How can any normal random variable $X \sim N(\mu, \sigma^2)$ be related to the standard normal random variable?
- ☐ How do we solve problems concerning the probability $P(X < a) = p$ for a normal random variable $X \sim N(\mu, \sigma^2)$? In particular, how do we find
 - ☐ the value of a or a related probability, given the values of a , μ and σ ;
 - ☐ the value of a given the values of p , μ and σ ;
 - ☐ the relationship between a , μ and σ , given the value of p ?
- ☐ How do we solve problems involving the use of $E(aX + bY)$ and $\text{Var}(aX + bY)$ given that X and Y are independent?
- ☐ How do we apply the normal distribution $N(\mu, \sigma^2)$ to solve statistical problems in real-world situations?

Key GC functions to master: normalcdf, invNorm.

2.1 Definition & Notation

What is a normal random variable?

(Knowledge of this function is not required for A-Levels.)

Definition 2.1.1 (Normal Random Variable)

A continuous random variable X whose probability density function can be expressed in the form

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

for some scalar μ and positive scalar σ is called a *normal random variable*.

Result 2.1.2 (Mean & Variance of a Normal Random Variable)

For a normal random variable X with probability density function as stated in **Definition 2.1.1**, it can be shown that

$$E(X) = \mu \quad \text{and} \quad \text{Var}(X) = \sigma^2.$$

We denote the distribution of X in this case by

$$X \sim N(\mu, \sigma^2).$$

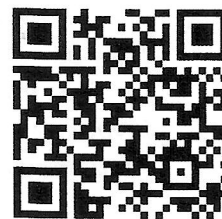
(Refer to [Appendix A](#) for a proof of **Result 2.1.2**).

2.2 The Normal Probability Density Curve

Access the link

<http://tinyurl.com/normaldistributioncurve>

or scan the QR code on the right and fill in the following table accordingly. (For the last two features, use the sliders to see the effect of changing the value of each parameter on the curve, keeping the other parameter constant.)



What are the features of a normal distribution curve?

How do the values of the mean μ and the standard deviation σ affect the shape and location of a normal distribution curve?

Feature	Description
Shape	Bell-shaped, symmetrical about the line $x = \mu$.
Asymptote	The x-axis is a horizontal asymptote of the curve.
Measures of Central Tendency	The mean, median and mode are all equal .
Effect of changing the value of μ	As μ increases, the normal distribution curve gets translated in the positive x-direction .
Effect of changing the value of σ	As σ increases, - the normal distribution curve becomes flatter and the maximum point becomes lower, - the area under the curve about the line $x = \mu$ becomes less concentrated, i.e. more dispersed.

2.3 Calculating Normal Probabilities and Inverse Normal Values

How do we use a GC to find probabilities involving a normal random variable?

Example 2.3.1

- (a) Given that $X \sim N(2, 4)$, find
- $P(3 < X \leq 10)$,
 - $P(X > 8)$.
- (b) Given that $Y \sim N(100, 36)$, find the values of a and b such that
- $P(Y \leq a) = 0.8907$,
 - $P(Y > b) = 0.1093$.

Solution:

(a)(i)

Since $P(X = a) = 0$,
 $P(3 < X \leq 10) =$
 $P(3 \leq X \leq 10)$.

	Press 2nd VAR 2 to call the 'normalcdf' function.
	Key in the values of the lower & upper bounds, as well as μ & σ , NOT σ^2 !
	Use the cursor buttons to select "Paste", then press the ENTER button twice.

$$P(3 < X \leq 10) = 0.309 \text{ (to 3 s.f.)}$$

(a)(ii)

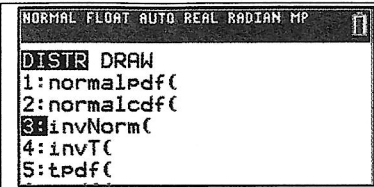
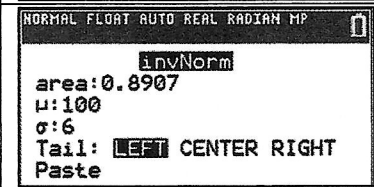
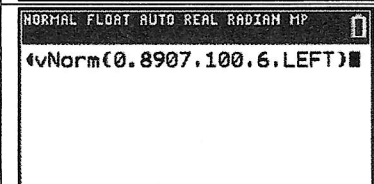
Since we cannot key in ∞ in the GC, we use a very large number instead such as E99, which is 10^{99} .

$$P(X > 8) = 0.0013499672$$

$$= 0.00135 \text{ (to 3 s.f.)}$$

(b)(i)

“invNorm” with LEFT is used for equations of the form $P(Y \leq a) = p$ or $P(Y < a) = p$.

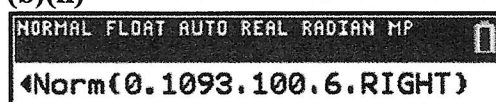
	Press    to call the ‘invNorm’ function.
	Key in the probability values in “area”, as well as μ & σ , NOT σ^2 !
	Use the cursor buttons to select “Paste”, then press the  button twice.

$$P(Y \leq a) = 0.8907 \Rightarrow a = 107 \text{ (to 3 s.f.)}$$

(b)(ii)

“invNorm” with RIGHT is used for equation of the form $P(Y \geq a) = p$ or $P(Y > a) = p$.

We may sometimes use the idea of the complement to manipulate given expressions. Here, $P(Y > b)$
 $= 1 - P(Y \leq b)$.



$$P(Y > b) = 0.1093 \Rightarrow b = 107 \text{ (to 3 s.f.)}$$

Alternatively,

$$P(Y > b) = 0.1093$$

$$\Rightarrow 1 - P(Y \leq b) = 0.1093$$

$$\Rightarrow P(Y \leq b) = 0.8907$$

$$\Rightarrow b = 107 \text{ (from part (i))}$$

2.4 Additive Properties of Normal Random Variables

Recall the properties of expectation and variance. For any random variables X and Y and real constants a and b , we have the following properties:

Expectation	Variance
$E(a) = a$	$\text{Var}(a) = 0$
$E(aX \pm b) = aE(X) \pm b$	$\text{Var}(aX \pm b) = a^2 \text{Var}(X)$
$E(aX \pm bY) = aE(X) \pm bE(Y)$	$\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$ if X and Y are independent.

It can be further shown that the sum of independent normal random variables are also normal random variables. Therefore, we have the following results:

If $X \sim N(\mu, \sigma^2)$ and a and b are real constants, then $aX + b \sim N(a\mu + b, a^2\sigma^2).$
If X and Y are independent normal random variables such that $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$, then $aX + bY \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2).$
If $X \sim N(\mu, \sigma^2)$, and $X_1, X_2, \dots, X_n$ are n independent observations of X, then $X_1 + X_2 + \dots + X_n \sim N(n\mu, n\sigma^2).$

How do we solve problems involving the use of $E(aX + bY)$ and $\text{Var}(aX + bY)$ given that X and Y are independent?

Example 2.4.1

Given that X and Y are independent random variables with distributions $N(100, 16)$ and $N(90, 20)$ respectively, find

- $P(X < Y)$,
- $P(X + 3Y > 385)$,
- $P(X + Y_1 + Y_2 + Y_3 > 400)$.

Solution:

How do we find the distribution of the sum or difference of two independent distinct normal random variables?

$$\begin{aligned}
 \text{(a)} \quad E(X - Y) &= E(X) - E(Y) \\
 &= 100 - 90 \\
 &= 10. \\
 \text{Var}(X - Y) &= \text{Var}(X) + \text{Var}(Y) \\
 &= 16 + 20 \\
 &= 36. \\
 \therefore X - Y &\sim N(10, 36). \\
 P(X < Y) &= P(X - Y < 0) \\
 &= 0.0478 \text{ (to 3 s.f.)}.
 \end{aligned}$$

How do we find the distribution of a scalar multiple of a normal random variable?

$$\begin{aligned}
 \text{(b)} \quad E(X + 3Y) &= E(X) + 3E(Y) \\
 &= 100 + 3 \times 90 \\
 &= 370. \\
 \text{Var}(X + 3Y) &= \text{Var}(X) + 3^2 \cdot \text{Var}(Y) \\
 &= 16 + 9 \times 20 \\
 &= 196. \\
 \therefore X + 3Y &\sim N(370, 196). \\
 P(X + 3Y > 385) &= 0.142 \text{ (to 3 s.f.)}.
 \end{aligned}$$

How do we find the distribution of sum of independent observations of a normal random variable?

$$\begin{aligned}
 \text{(c)} \quad E(X + Y_1 + Y_2 + Y_3) &= E(X) + 3E(Y) \\
 &= 100 + 3 \times 90 \\
 &= 370. \\
 \text{Var}(X + Y_1 + Y_2 + Y_3) &= \text{Var}(X) + 3\text{Var}(Y) \\
 &= 16 + 3 \times 20 \\
 &= 76. \\
 \therefore X + Y_1 + Y_2 + Y_3 &\sim N(370, 76). \\
 P(X + Y_1 + Y_2 + Y_3 > 400) &= 0.000290 \text{ (to 3 s.f.)}
 \end{aligned}$$

Note from (b) and (c) that $X + 3Y$ and $X + Y_1 + Y_2 + Y_3$ do NOT have the same distribution!

How do we apply the normal distribution $N(\mu, \sigma^2)$ to solve statistical problems in real-world situations?

How do we solve problems involving the use of $E(aX + bY)$ and $\text{Var}(aX + bY)$ given that X and Y are independent?

Define the random variables given in the question and state their distribution.

Example 2.4.2

The length, in cm, of a rectangular tile is a normal variable with mean 19.8 and standard deviation 0.1. The breadth, in cm, is another independent normal variable with mean 9.8 and standard deviation 0.1. Find the probability that

- (i) the sum of the lengths of five randomly chosen tiles exceeds 99.4 cm,
- (ii) the breadth of a randomly chosen tile is less than half of its length,
- (iii) the sum of the lengths of 50 randomly chosen tiles exceeds the sum of the breadths of 90 randomly chosen tiles by more than 109 cm.

Solution:

Let L and B denote the length and breadth of a randomly chosen rectangular tile, in cm, respectively. We have

$$L \sim N(19.8, 0.1^2) \text{ and } B \sim N(9.8, 0.1^2).$$

- (i) Let L_1, L_2, L_3, L_4 and L_5 denote the lengths of 5 randomly chosen tiles, in cm. Then,

$$\begin{aligned}
 E(L_1 + L_2 + L_3 + L_4 + L_5) &= 5E(L) \\
 &= 5 \times 19.8 \\
 &= 99
 \end{aligned}$$

For (i), (ii) and (iii), we

- define/introduce a new random variable,
- find its mean and variance,
- state its distribution, and finally
- compute the required probabilities.

and

$$\begin{aligned}\text{Var}(L_1 + L_2 + L_3 + L_4 + L_5) &= 5\text{Var}(L) \\ &= 5 \times 0.1^2 \\ &= 0.05.\end{aligned}$$

So $L_1 + L_2 + L_3 + L_4 + L_5 \sim N(99, 0.05)$.

Therefore,

$$P(L_1 + L_2 + L_3 + L_4 + L_5 > 99.4) = 0.0368 \text{ (to 3 s.f.)}.$$

(ii)

$$\begin{aligned}E\left(B - \frac{1}{2}L\right) &= E(B) - \frac{1}{2}E(L) \\ &= 9.8 - \frac{19.8}{2} \\ &= -0.1\end{aligned}$$

and

$$\begin{aligned}\text{Var}\left(B - \frac{1}{2}L\right) &= \text{Var}(B) + \left(\frac{1}{2}\right)^2 \text{Var}(L) \\ &= 0.1^2 + \frac{0.1^2}{4} \\ &= 0.0125.\end{aligned}$$

So $B - \frac{1}{2}L \sim N(-0.1, 0.0125)$.

Therefore,

$$\begin{aligned}P\left(B < \frac{1}{2}L\right) &= P\left(B - \frac{1}{2}L < 0\right) \\ &= 0.814 \text{ (to 3 s.f.)}.\end{aligned}$$

(iii) Let $D = L_1 + L_2 + \dots + L_{50} - (B_1 + B_2 + \dots + B_{90})$.

Then

$$\begin{aligned}E(D) &= 50E(L) - 90E(B) \\ &= 50 \times 19.8 - 90 \times 9.8 \\ &= 108\end{aligned}$$

and

$$\begin{aligned}\text{Var}(D) &= 50\text{Var}(L) + 90\text{Var}(B) \\ &= 50 \times 0.01 + 90 \times 0.01 \\ &= 1.4.\end{aligned}$$

So $D \sim N(108, 1.4)$.

For the probability that the sum of the lengths of the 50 tiles exceeds the sum of the breadths of the 90 tiles by more than 109 cm, we have

$$P(D > 109) = 0.199 \text{ (to 3 s.f.)}.$$

How do we apply the normal distribution $N(\mu, \sigma^2)$ to solve statistical problems in real-world situations?

Exercise 2.4.3

A manufacturer produces specialist light bulbs. The masses in grams of one type of light bulb have the normal distribution $N(50, 1.5^2)$. Each light bulb is packed into a randomly chosen box. The masses in grams of the empty boxes have distribution $N(75, 2^2)$.

- (i) Sketch the distribution for the masses of specialist light bulbs between 40 grams and 60 grams.
- (ii) Find the probability that the mass of a randomly chosen bulb is less than 50.4 grams.
- (iii) Find the probability that the mass of a randomly chosen bulb is between 50500 and 50650 milligrams.
- (iv) If one light bulb is chosen at random, find the probability that its mass differs from the mean mass by at least 0.5 grams.
- (v) Find the value of a such that 20% of the light bulbs have mass of at least a grams.
- (vi) Find the probability that the total mass of 4 randomly chosen empty boxes is more than 297 grams.
- (vii) Find the probability that 4 times the mass of a randomly chosen empty box is more than 297 grams.
- (viii) Find the probability that the total mass of a randomly chosen bulb and a randomly chosen empty box is between 124.9 and 125.7 grams. State a necessary assumption for the calculations in (viii) to be valid.

How do we solve problems involving the use of $E(aX + bY)$ and $\text{Var}(aX + bY)$ given that X and Y are independent?

- (ix) The manufacturing cost of the specialist light bulbs and the empty boxes is \$0.02 per gram and \$0.01 per gram respectively. Let V and W represent the manufacturing cost of one specialist light bulb and one empty box respectively. Find $P(100V - 90W < 30)$ and explain, in the context of this question, what your answer represents.

[2018 GCE A Level P2 Q10 (modified)]

Solution:

2.5 Standard Normal Distribution

What is the standard normal random variable?

Definition 2.5.1 (Standard Normal Random Variable)

A normal random variable Z with mean 0 and variance 1, i.e.

$$Z \sim N(0, 1),$$

is called the *standard* normal random variable and is said to follow the *standard* normal distribution.

How do we standardise a normal random variable?

Result 2.5.2 (Standardisation)

If $X \sim N(\mu, \sigma^2)$, then the random variable defined by

$$Z = \frac{X - \mu}{\sigma}$$

is the standard normal random variable.

Proof of Result 2.5.2:

Suppose $X \sim N(\mu, \sigma^2)$

and Z is defined as

$$Z = \frac{X - \mu}{\sigma}.$$

$$E(Z)$$

$$= E\left(\frac{X - \mu}{\sigma}\right)$$

$$= \frac{1}{\sigma} E(X - \mu)$$

$$= \frac{1}{\sigma} [E(X) - E(\mu)]$$

$$= \frac{1}{\sigma} (\mu - \mu)$$

$$= 0$$

$$\text{Var}(Z)$$

$$= \text{Var}\left(\frac{X - \mu}{\sigma}\right)$$

$$= \frac{1}{\sigma^2} \text{Var}(X - \mu)$$

$$= \frac{1}{\sigma^2} [\text{Var}(X) + \text{Var}(\mu)]$$

$$= \frac{1}{\sigma^2} \text{Var}(X)$$

$$= \frac{1}{\sigma^2} \cdot \sigma^2 = 1.$$

By the additive property of normal random variables,

$$Z \sim N(0, 1).$$

Big Idea: Transformation

The process of constructing the standard normal random variable by subtracting a given normal random variable by its mean, followed by dividing by its standard deviation is called *standardisation*, and this can be seen as a transformation that not only preserves the shape of the original normal distribution but simplifies probability calculations by reducing all normal distributions to a common scale.

This transformation is useful in solving a wide range of practical problems where:

- the mean or/and standard deviation of the normal random variable is/are unknown,
- we need to make use of the symmetry of the normal distribution about its mean,
- we can easily compare probability values (without computing them) from different normal distributions with different means and variances,
- we need to make statistical inferences such as in hypothesis testing (which we will learn about in topic 19).

How do we find a probability involving a normal random variable or an inverse normal value by considering the symmetry of a normal distribution curve about its mean?

Example 2.5.3

If $Z \sim N(0, 1)$, find

- (i) $P(Z > 0)$,
- (ii) $P(|Z| \leq 1.5)$,
- (iii) the value of a such that $P(|Z| \leq a) = 0.95$.

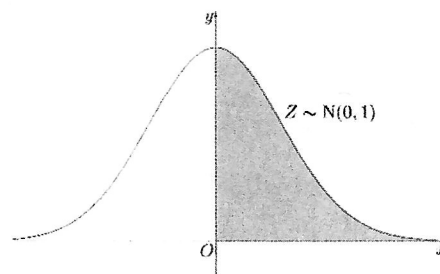
The continuous random variable X has the normal distribution $X \sim N(7, 35)$. Find the value of b such that

$$P(|X - 7| \leq b) = 0.95.$$

Solution:

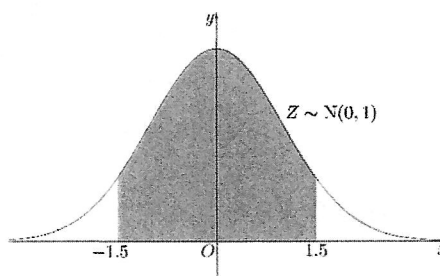
It is generally useful to first sketch a relevant curve and shaded region to provide a visualisation to the problem.

(i)



$$P(Z > 0) = 0.5 \text{ (by symmetry).}$$

(ii)

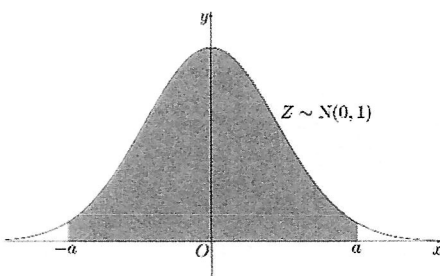


$$\begin{aligned} P(|Z| \leq 1.5) &= P(-1.5 \leq Z \leq 1.5) \\ &= 0.866. \end{aligned}$$

Pause and Think:

- Does $|Z|$ follow a normal distribution?
- Why do we need to resolve the modulus sign in part (ii)?

(iii)

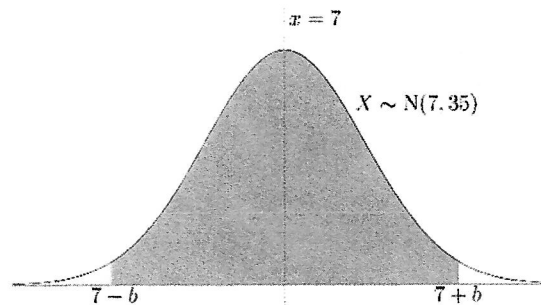


$$\begin{aligned} P(|Z| \leq a) &= 0.95 \Rightarrow P(-a \leq Z \leq a) = 0.95 \\ &\Rightarrow P(Z \leq -a) = \frac{1 - 0.95}{2} \\ &= 0.025 \\ &\Rightarrow -a = -1.9600 \\ &\Rightarrow a = 1.96 \text{ (to 3 s.f.).} \end{aligned}$$

(Last Part)

Method 1: Applying Symmetry

Since the probability density curve of X is symmetrical about the line $x = 7$ ($\because E(X) = 7$),



$$P(|X - 7| \leq b) = 0.95$$

$$P(-b \leq X - 7 \leq b) = 0.95$$

$$P(7 - b \leq X \leq 7 + b) = 0.95$$

$$1 - 2P(X \leq 7 - b) = 0.95$$

$$P(X \leq 7 - b) = \frac{1 - 0.95}{2}$$

$$= 0.025$$

$$7 - b = -4.5953$$

$$b = 7 + 4.5953$$

$$= 11.6 \text{ (to 3 s.f.)}$$

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NORMAL FLOAT AUTO REAL RADIAN MP
invNorm
area:0.025
μ:7
σ:√(35)
Tail: LEFT CENTER RIGHT
Paste

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Method 2: Using Standardisation

Since $X \sim N(7, 35)$, $Z = \frac{X - 7}{\sqrt{35}} \sim N(0, 1)$. Therefore,

$$P(|X - 7| \leq b) = 0.95$$

$$P\left(\left|\frac{X - 7}{\sqrt{35}}\right| \leq \frac{b}{\sqrt{35}}\right) = 0.95$$

$$P\left(|Z| \leq \frac{b}{\sqrt{35}}\right) = 0.95.$$

From the result in part (iii), $\frac{b}{\sqrt{35}} = 1.9600$. Therefore,

$$b = 1.9600\sqrt{35}$$

$$= 11.596$$

$$= 11.6 \text{ (to 3 s.f.)}$$

How can any normal random variable $X \sim N(\mu, \sigma^2)$ be related to the standard normal random variable?

Define the random variable given in the question and state its distribution.

Use standardisation so that we can work with a distribution with known parameters, in this case, $N(0, 1)$.

Example 2.5.4

Find the percentages of the observations of a normal random variable X that lie within 1, 2 and 3 standard deviations from its mean respectively.

Solution:

Let $X \sim N(\mu, \sigma^2)$. Then $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$.

Proportion of observations of X that lie within 1 standard deviation from its mean,

$$\begin{aligned} P(|X - \mu| \leq \sigma) &= P\left(\left|\frac{X - \mu}{\sigma}\right| \leq 1\right) \\ &= P(|Z| \leq 1) \\ &= P(-1 \leq Z \leq 1) \\ &= 0.68269 \\ &= 0.683 \text{ (to 3 s.f.)}. \end{aligned}$$

Proportion of observations of X that lie within 2 standard deviations from its mean,

$$\begin{aligned} P(|X - \mu| \leq 2\sigma) &= P\left(\left|\frac{X - \mu}{\sigma}\right| \leq 2\right) \\ &= P(|Z| \leq 2) \\ &= P(-2 \leq Z \leq 2) \\ &= 0.95499876 \\ &= 0.954 \text{ (to 3 s.f.)}. \end{aligned}$$

Proportion of observations of X that lie within 3 standard deviations from its mean,

$$\begin{aligned} P(|X - \mu| \leq 3\sigma) &= P\left(\left|\frac{X - \mu}{\sigma}\right| \leq 3\right) \\ &= P(|Z| \leq 3) \\ &= P(-3 \leq Z \leq 3) \\ &= 0.99730 \\ &= 0.997 \text{ (to 3 s.f.)}. \end{aligned}$$

Therefore, approximately 68.3%, 95.4% and 99.7% of all observations of X lie within 1, 2 and 3 standard deviations from the mean of X respectively.

How do we find an unknown normal distribution parameter (or parameters) by using standardisation?

Example 2.5.5

The heights, in metres, of a certain population of adult males follow a normal distribution with mean μ and variance σ^2 . If 10% of the population have heights greater than 1.8 m and 5% have heights less than 1.6 m, find μ and σ .

Solution:

Define the random variables given in the question and state their distribution.

Let X denote the height of a randomly chosen adult male from this population, in metres.

Then $X \sim N(\mu, \sigma^2)$.

Since 10% of the population have heights greater than 1.8 m, $P(X > 1.8) = 0.10$.

Use standardisation to obtain an equation involving μ and σ .

$$P\left(Z > \frac{1.8 - \mu}{\sigma}\right) = 0.10$$

$$1 - P\left(Z \leq \frac{1.8 - \mu}{\sigma}\right) = 0.10 \quad \text{--- (*)}$$

$$P\left(Z \leq \frac{1.8 - \mu}{\sigma}\right) = 0.90 \quad \text{--- (*)}$$

$$\frac{1.8 - \mu}{\sigma} = 1.2816. \quad \text{--- (1)}$$

Since 5% of the population have heights less than 1.6 m, $P(X < 1.6) = 0.05$.

Use standardisation to obtain a second equation involving μ and σ .

$$P\left(Z < \frac{1.6 - \mu}{\sigma}\right) = 0.05$$

$$\frac{1.6 - \mu}{\sigma} = -1.6449. \quad \text{--- (2)}$$

Using GC to solve (1) and (2) simultaneously, we get $\mu = 1.71$ and $\sigma = 0.0683$.

(*) is not required if there is an option of choosing RIGHT tail under the invNorm command.

Summary of Solving Problems Involving Normal Distribution Probabilities

Problem Form			
$P(X \leq a) = p, \text{-----} (*)$ where $X \sim N(\mu, \sigma^2)$.			
Quantities Given	Quantities to be Found	Reference Example	Steps Taken
a, μ and σ^2	p	Example 2.3.1(a)	Use the “normalcdf” function in the GC.
p, μ and σ^2	a	Example 2.3.1(b)	Use the “invNorm” function in the GC.
a and p	μ and/or σ^2	Example 2.5.5	(If the value of μ is unknown, first check if we can use symmetry to find the value of μ .) (1) Standardise the random variable, i.e. $Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$ (2) Rewrite (*) in terms of Z, i.e. $P\left(Z \leq \frac{a - \mu}{\sigma}\right) = p.$ (3) Use the “invNorm” function in the GC to find the value of $\frac{a - \mu}{\sigma}$. (4) Solve for μ and/or σ^2 by using the equation(s) obtained from (3).

Appendix A (Proof of Result 2.1.2)

For any continuous random variable X with probability density function f , its mean and variance are given by

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

and

$$\text{Var}(X) = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx.$$

For a normal distribution with mean μ and standard deviation σ , its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}.$$

So,

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \right] dx \\ &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\ &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\mu - \sigma^2 \left(-\frac{x-\mu}{\sigma^2} \right) \right] e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\ &= \mu \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx - \frac{\sigma}{\sqrt{2\pi}} \left[e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \right]_{-\infty}^{\infty} \\ &= \mu(1) - \frac{\sigma}{\sqrt{2\pi}}(0-0) \\ &= \mu \end{aligned}$$

and

$$\begin{aligned} \text{Var}(X) &= E[(X - \mu)^2] \\ &= \int_{-\infty}^{\infty} (x - \mu)^2 \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \right] dx \\ &= \left[-\frac{\sigma(x-\mu)}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} -\frac{\sigma}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\ &= (0-0) + \sigma^2 \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\ &= \sigma^2. \end{aligned}$$